Mind Reading Using Machine Learning and EEG

Technical report, team AI Toddlers · Livien Majer, Sebastian Leitl, Daniel Daenecke, Sofiya Asadchaya · National AI Hackathon 2026 (SchubKI), October 2026
Code, weights and predictions: **https://github.com/LivienMajer/EEG-Mindreading** (branch `main`; data analysis and RSA on branch `feature/eeg-preprocessing`)

Summary. We decode 3-second EEG recordings into the representation spaces of frozen, public video and audio models and rank the 102 candidate videos by similarity. Seven decoders ("votes") are summed and re-ranked jointly with Sinkhorn normalisation. The targets were chosen with a brain-vs-AI representational similarity analysis (RSA): mid-level layers of AlexNet and VideoMAE, and the loudness of the sound track, are the most brain-like, and more brain-like targets decode better. On the four development subjects, with every setting chosen by leave-one-subject-out, the final model reaches **44.9 % top-1 and 76.7 % top-5** (mean rank 4.36 of 102; chance 1.0 % and 4.9 %; organizers' baseline 3.9 % / 11.8 %).

1. Task and data

For each test recording, the 102 candidate videos must be ranked from most to least likely to have caused it; performance is top-1 and top-5 accuracy. Six subjects watched 3 s videos with sound. Each subject has 850–972 training recordings (one per video, 1,000 training videos in total) and 102 test recordings of the same 102 test videos, one each, in shuffled order. Test labels are given for the development subjects sub-01..04 and withheld for sub-05/06. The EEG has 17 posterior channels (Pz P3 P7 O1 Oz O2 P4 P8 P1 P5 PO7 PO3 POz PO4 PO8 P6 P2), 500 Hz, 1500 samples from video onset, pre-processed by the organizers (0.1–40 Hz band-pass, blink ICA, 1000 → 500 Hz). Test data are never used for training; dev test labels are used only to choose settings, as the rules allow.

Our approach builds on encoding and decoding models of brain activity (Naselaris et al. 2011) and on recent work that decodes EEG and MEG into the embedding spaces of pretrained networks with contrastive learning (Défossez et al. 2023; Song et al. 2024; Benchetrit et al. 2024). Unlike large image-EEG datasets such as THINGS-EEG2 (Gifford et al. 2022), this challenge offers one short recording per video and subject from 17 posterior channels.

2. Data analysis and preprocessing

2.1 Automatic quality control

The package `eeg_pipeline` computes quality metrics per subject (integrity, flat or clipped channels, artefact trials, line noise, existing filtering, trial offsets, test-vs-train amplitude) and turns them into one shared preprocessing configuration with fixed, documented rules; test labels are never used. On this dataset: no channel or trial was dropped, no notch was needed, trial offsets were as large as the within-trial signal (per-trial demeaning), and test recordings had 0.60–0.70× the amplitude of training recordings in every subject (normalisation per split).

2.2 Findings that shaped the model

- **High frequencies carry signal.** Strong 30–50 Hz power is locked to the video. Band-pass, notch, average reference and Laplacian filtering all made videos harder to tell apart, so the model uses only a 40 Hz low-pass and no average reference (which would also remove the shared posterior signal).
- **The mean response was removed upstream**: train and test average responses anti-correlate ($r = -0.71$ to -0.92). Each split's mean response is therefore removed, and average-response (ERP) plots are not interpretable.
- **Test recordings behave like averages of about two repetitions.** The correlation between two subjects' responses to the same video is 2.17× higher in test than in training (95 % CI 1.85–2.48 over 6 subject pairs; averaging two repetitions predicts 2.00, a pure change of scale 1.00), and power

drops less at low than at high frequencies (implied k = 1.8–2.4). This explains the lower test amplitude, why validation on single training trials underestimates test scores, and why the ridge decoders need less regularisation at test time (Section 4).

- **Every candidate appears exactly once per subject's test set**, which the scoring step exploits (Sinkhorn, Section 4).
- **The information is in the onset response, 50–600 ms.** Decoding the frame shown at 1 s or 2 s from the EEG 50–600 ms later is near chance (mean rank 40–43 of 102), while the first frame alone is as good a target as the whole video.
- **Channel names match the file rows**: channel correlation falls with scalp distance (rho = 0.93; random orders ≤ 0.45, p = 0.0002), so spatial results are valid.

Preprocessing of the final model (label-free, per subject and split): 40 Hz low-pass, 0–1 s, resampling to 100 Hz, removal of the split's mean response, scaling, clipping at ±6 SD and Euclidean alignment (He & Wu 2020). The per-subject InfoNCE networks use per-trial channel z-scoring at 100 Hz, 50–350 ms.

3. Brain vs AI: representational similarity analysis

Method. RSA (Kriegeskorte et al. 2008), resolved in time as in Cichy et al. (2014, 2016), on the training recordings of the 695 videos seen by all six subjects (no test data). For each subject and each 50 ms window (every 20 ms, 0–800 ms), a brain dissimilarity matrix is computed from the pattern over 17 channels × 5 time bins (baseline 0–50 ms; 10 ms bin averages instead of a filter, which could shift effects in time (Widmann et al. 2015); subject's mean response removed). It is compared by Spearman correlation with the dissimilarity matrices of 30 feature sets: simple visual properties and loudness over time, AlexNet conv1–fc7, CLIP ViT-B/32 blocks 3–12 and output, VideoMAE layers 4/8/12, V-JEPA 2 layers 6–24 (Assran et al. 2025), and CLAP. Statistics: lower noise ceiling (Nili et al. 2014), one-sided t-tests over subjects with FDR correction (Benjamini & Hochberg 1995), jackknife latency tests (Miller et al. 1998), electrode groups and a rerun without the 26 silent videos.

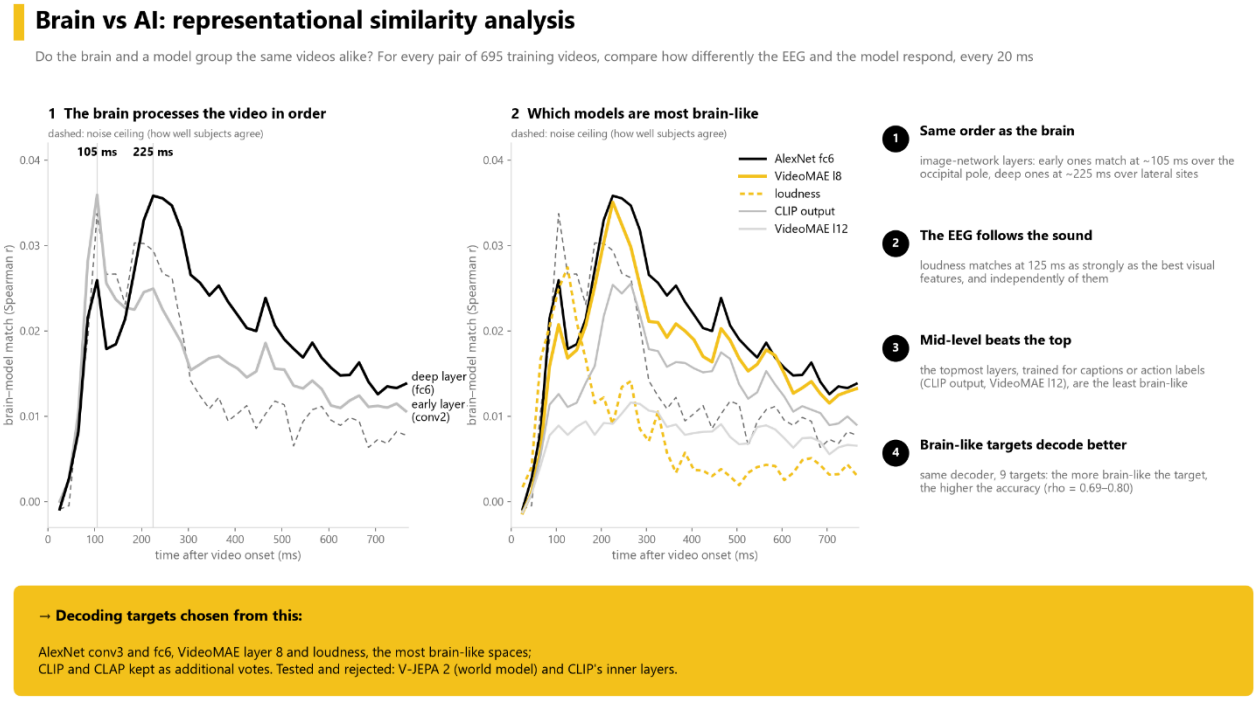


Figure 1. Brain–model match over time. Left: an early AlexNet layer peaks at 105 ms, a deep layer at 225 ms. Right: AlexNet fc6, VideoMAE layer 8 and loudness are the most brain-like; CLIP's output and VideoMAE's top layer are weaker. Dashed: noise ceiling.

- **The brain processes the video in order.** AlexNet conv1–3 match best at 105 ms over the occipital pole; conv4–fc7 at 225 ms over lateral parieto-occipital sites (fc6: r = 0.036, 122 % of the noise ceiling). Early layers win the first wave and deep layers the second in 6/6 subjects (t(5) = 5.60, p =

0.001); the peak shifts by 120 ms (jackknife t(5) = 5.69, p = 0.001). This matches the hierarchical correspondence between deep networks and the visual system (Güçlü & van Gerven 2015; Yamins & DiCarlo 2016; Cichy et al. 2016); the lateral sites lie over occipito-temporal cortex, where objects and scenes are processed (Grill-Spector & Malach 2004).

- **Mid-level layers are the most brain-like** (VideoMAE layer 8: 0.035). The topmost layers, trained for captions or action labels, are the least brain-like (CLIP output 0.026, VideoMAE layer 12 0.012). CLIP's inner blocks (0.032) are better than its output. The V-JEPA 2 world model peaks at 0.030 (layer 18).
- **The EEG follows the sound.** Loudness over time matches as strongly as the best visual features (0.027 at 125 ms, the latency of the auditory N1; Näätänen & Picton 1987) and is nearly independent of every visual model (overlap r ≤ 0.07).
- **Brain-likeness predicts decoding.** The same per-subject decoder trained toward 9 of these targets decodes better the more brain-like the target is, in the spirit of benchmarking networks by their brain similarity (Schrimpf et al. 2018) (Spearman rho = 0.69, p = 0.038; 0.80, p = 0.017 without the 30-dimensional loudness target). Checked without any test data on 102 hidden training videos per subject in all six subjects: top-1 / top-5 CLIP 4.7 / 18.8 %, AlexNet fc6 6.2 / 22.5 %, VideoMAE layer 8 6.9 / 22.8 % (rank $p \leq 2 \times 10^{-11}$ vs CLIP).

Consequence: the decoders target AlexNet conv3/fc6, VideoMAE layer 8 and loudness; CLIP and CLAP are kept as additional votes.

4. Model

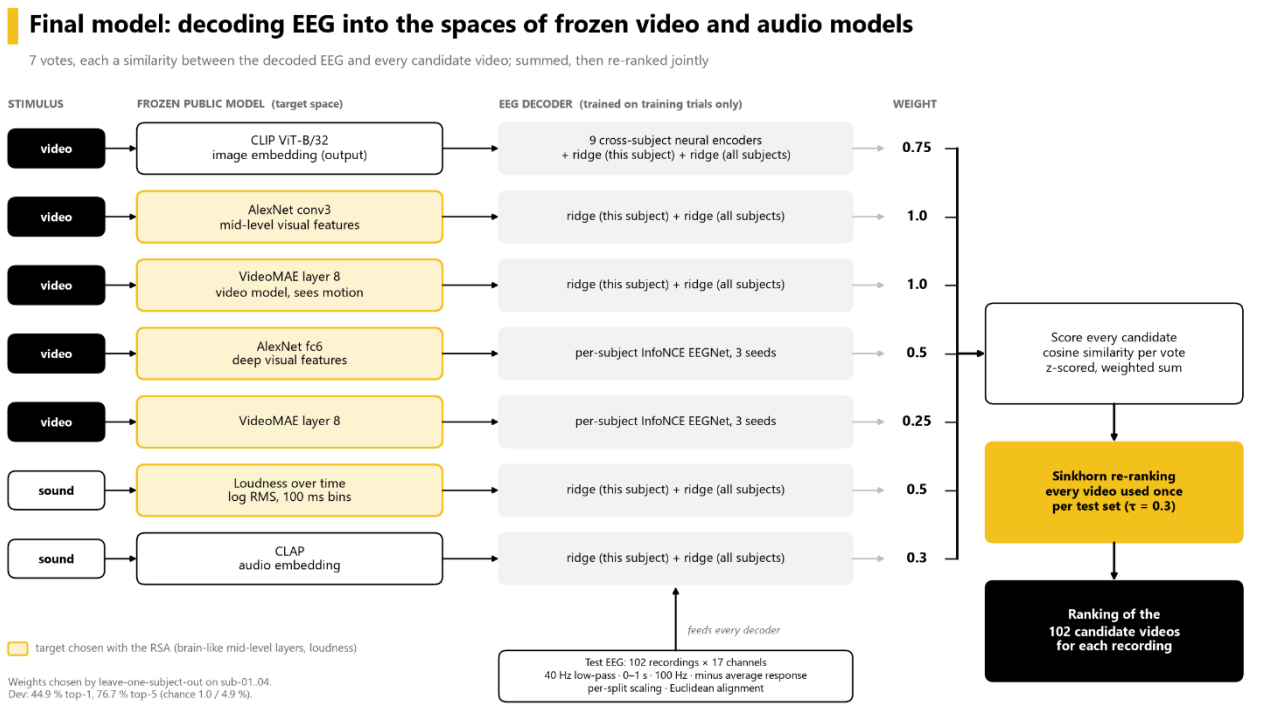


Figure 2. Final model. Seven votes, each the similarity between the decoded EEG and every candidate in one frozen model's space, are summed and re-ranked with Sinkhorn. Yellow: targets chosen with the RSA.

4.1 Target spaces (frozen public models)

Target	Checkpoint	Feature
CLIP image embedding	OpenCLIP ViT-B-32, `openai` (open_clip 3.3.0); Radford et al. 2021, Cherti et al. 2023	output, L2-normalised, mean of 8 frames

AlexNet conv3 / fc6	torchvision AlexNet `IMAGENET1K_V1`; Krizhevsky et al. 2017	conv3 after ReLU pooled to 4×4, fc6 after ReLU; mean of 8 frames
VideoMAE layer 8	`MCG-NJU/videomae-base-finetuned-kinetics`; Tong et al. 2022	16 frames as one clip, tokens averaged
CLAP audio	`laion/clap-htsat-unfused`; Wu et al. 2023	3 s sound track
Loudness	none	log RMS of the sound track, 100 ms bins (30 values)

4.2 EEG decoders (trained on training recordings only)

- **Cross-subject neural encoders** (CLIP vote): 5 ShallowConvNet (Schirrmeister et al. 2017), 3 MLP and 1 EEGNet (Lawhern et al. 2018), each with a per-subject channel-mixing layer, similar to the subject layer of Défossez et al. (2023), so that one network serves all six subjects; input 50–600 ms. Trained on all subjects' training recordings with InfoNCE (van den Oord et al. 2018) over all training videos (learnable temperature, initial 0.07) plus 0.5 × (1 − cosine) toward the target, 150 epochs with early stopping on 10 % held-out videos, and augmentation (time shift, channel dropout, noise, scaling).
- **Ridge decoders** (Hoerl & Kennard 1970; every vote except the InfoNCE ones): EEG amplitude of every 10 ms sample from 50 to 600 ms on 17 channels (935 features) regressed onto the target. One ridge per subject plus one pooled over all subjects (weight 0.25). The regularisation is the cross-validated choice × 0.2, because cross-validation runs on single training trials while the test recordings are averages of about two (Section 2.2).
- **Per-subject InfoNCE EEGNets** (fc6 and VideoMAE-l8 votes): one small EEGNet per subject (temporal convolution, depthwise spatial and separable convolution, projection to 256 dimensions) maps 50–350 ms into a 256-dimensional PCA of the target, fitted on the 1,000 training videos. Symmetric InfoNCE (temperature 0.1, AdamW 3×10^{-4}, cosine schedule, 200 epochs); three seeds averaged.

4.3 Scoring

For each vote, the cosine similarity between the decoded EEG and each candidate's embedding gives a 102 × 102 table; each table is z-scored globally and the tables are summed with the weights CLIP 0.75, conv3 1.0, VideoMAE-l8 ridge 1.0, CLAP 0.3, loudness 0.5, InfoNCE fc6 0.5, InfoNCE VideoMAE-l8 0.25. **Sinkhorn re-ranking** (Sinkhorn 1964; Cuturi 2013) then uses that every candidate appears once per test set: the summed score S is turned into $P = \exp(S/\tau)$, $\tau = 0.3$, and rows and columns of P are normalised in turn until each recording and each candidate sums to one; each recording is ranked by its row. This counters hubness, the tendency of a few candidates to be the nearest neighbour of many queries in high-dimensional spaces (Radovanović et al. 2010; Dinu et al. 2015), which here would let "attractive" videos top many recordings. It uses no labels and counts as unsupervised normalisation of the test recordings.

5. Validation protocol

- Models are trained on the training recordings of all six subjects only.
- Every weight and setting is chosen by leave-one-subject-out on sub-01..04: chosen on three subjects, scored on the fourth, four times. For the final weights all four folds picked the same values, so the held-out estimate equals the in-sample value.
- A change is kept only if it wins on the same 408 recordings (exact McNemar test for hits, McNemar 1947; Wilcoxon signed-rank test for the rank of the true video, Wilcoxon 1945).
- The RSA-chosen targets were confirmed without test data on hidden training videos (Section 3).

6. Results

Results on the development subjects

sub-01..04: 408 test recordings, each ranked against 102 candidate videos; final model's settings all chosen by leave-one-subject-out

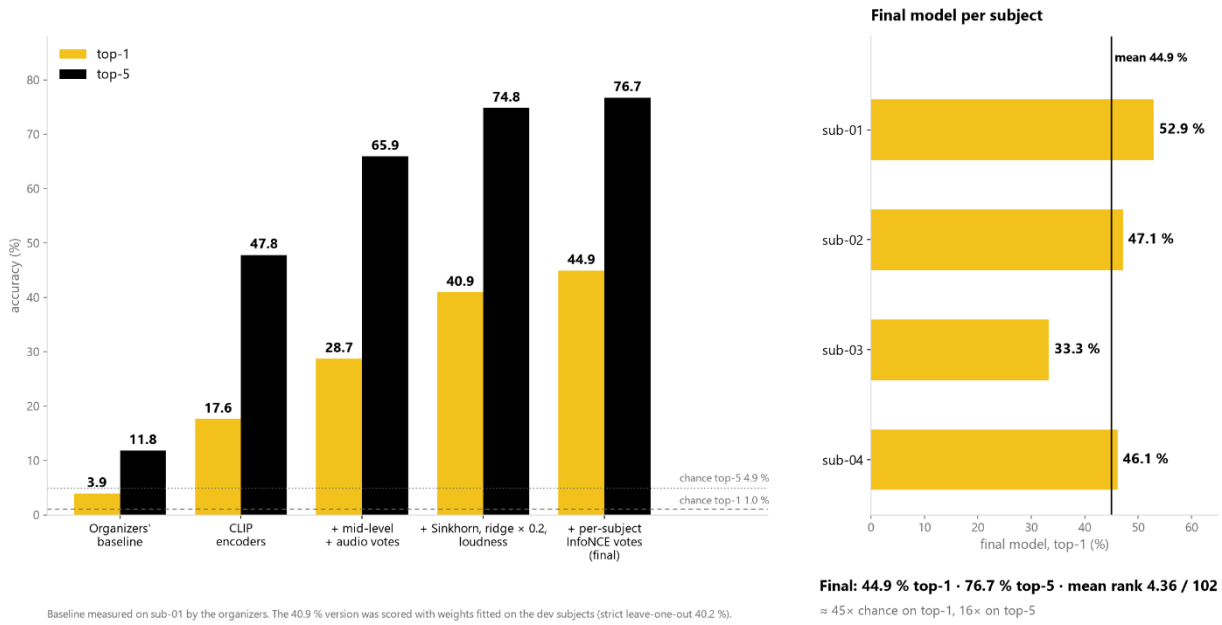


Figure 3. Development subjects. Left: top-1 and top-5 along the development of the model. Right: final top-1 per subject.

Configuration (dev, sub-01..04)	top-1 %	top-5 %	mean rank
Final model	**44.9**	**76.7**	**4.36**
without Sinkhorn	28.9	61.0	6.96
without AlexNet conv3 vote	39.0	73.5	4.98
without CLAP vote	41.2	76.2	4.56
without CLIP image ensemble	41.7	71.6	4.93
without InfoNCE VideoMAE-l8 vote	41.9	75.7	4.50
without InfoNCE fc6 vote	42.2	76.0	4.65
without loudness vote	42.6	74.5	4.69
without VideoMAE-l8 ridge vote	42.9	76.5	4.44
organizers' baseline (AlexNet forward model, sub-01)	3.9	11.8	–
chance	1.0	4.9	51.5

Table 1. Ablation of the final model: each row removes one component. Per subject (top-1): 52.9 / 47.1 / 33.3 / 46.1 %.

Every vote contributes 2–6 points of top-1; Sinkhorn contributes 16. A single ridge into AlexNet conv3 or VideoMAE-l8 space alone (25.0 % top-1) beats the whole CLIP ensemble alone (21.1 %), as the RSA predicts.

Reducing the ridge regularisation (×0.2) was worth 4.2 points in the previous version of the model (40.9 % vs 36.8 %). Sub-03, the subject with the weakest stimulus-locked signal in its training data, is the hardest in almost every configuration.

7. What did not work

All tested with the same leave-one-subject-out protocol and paired tests; none gave a significant gain.

- **Other target models:** V-JEPA 2 (less brain-like than AlexNet and VideoMAE); CLIP's inner layers (better than its output, not better than AlexNet fc6); video-to-text captions (text is further from what posterior EEG carries).
- **Other parts of the signal:** EEG after 600 ms toward the frame shown at that time (near chance); early EEG toward early layers only (worse: every layer carries information in both waves); band power and covariance features (no information); average reference, high-pass, spectral whitening (remove stimulus information).
- **Other fusion and adaptation:** more members, cross-subject transfers, score instead of rank fusion (saturated); a hard one-to-one assignment (Hungarian algorithm; Kuhn 1955) instead of Sinkhorn; the loudness envelope as a stand-alone decoder (too coarse to pick one of 102 videos; useful only as a vote); cross-subject pseudo-averages, test-time augmentation; soft or weighted contrastive targets and joint EEG/video/audio alignment for the InfoNCE votes (no gain or worse).

Lesson: the gains came from choosing brain-like targets and from using the structure of the test set, not from adding models or more of the signal.

8. Limitations

- Only 17 posterior electrodes: frontal, central and temporal responses are not recorded.
- One noisy recording per training video and subject; test recordings are averages, so train and test differ.
- Small evaluation: 408 development and 204 held-out recordings, so differences of 2–3 points are noise. Settings were chosen on the development subjects, and the held-out subjects are expected to score lower (sub-06 also has the fewest training recordings, 850).
- Sinkhorn uses a property of this benchmark (each candidate once per test set); an open-ended setting that sees one recording at a time could not use it.
- What is decoded is the visual and sound "fingerprint" of the first 600 ms after onset, not the meaning or the story of the video.

9. Reproducibility

Python 3.11; `pip install -r requirements.txt`; with the organizers' data in `data/eeg-mind-reading-challenge/`, `python -m src.xsubj.submit` rebuilds `submissions/sub-05_ranked_candidate_list.npy` and `sub-06_…` from the committed weights and features in a few minutes on CPU and scores sub-01..04. All encoder, ridge and InfoNCE weights and all video and audio features are versioned in the repository; public models are named by exact checkpoint (Section 4.1). Retraining commands are listed in `src/xsubj/README.md`; the feature extractor for AlexNet, VideoMAE and loudness is `eeg_pipeline/rsa_features.py`. The data analysis of Sections 2–3 (quality control and preprocessing package, test-averaging and channel-order checks, RSA, with all results and commands) is on branch `feature/eeg-preprocessing` of the same repository (`eeg_pipeline/`, `results/rsa/`, `eeg_pipeline/README.md`). Detailed reports: `docs/report_eeg_encoding.md` and `docs/report_infonce_votes.md` on `main`.

10. Code guide

The submission is produced by one entry point, `python -m src.xsubj.submit`, and every setting is read from one file, `configs/xsubj.yaml`. All code of the final model is in `src/xsubj/`; it imports nothing from the other folders.

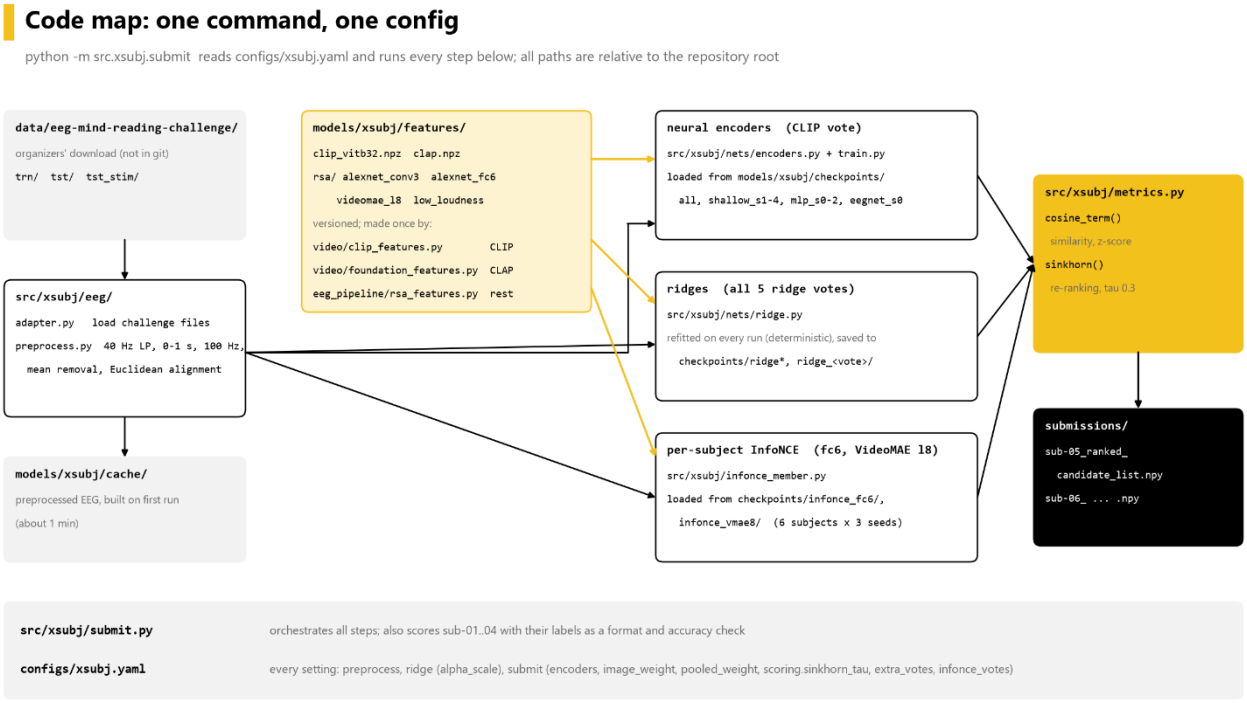


Figure 4. Code map of the final model on branch `main`: which file implements each step, and where data, features and weights live.

10.1 Pipeline step → code

Step (report section)	Code	Data / weights
Load the challenge files (1)	`src/xsubj/eeg/adapter.py`	`data/eeg-mind-reading-challenge/` (not in git)
Preprocessing (2)	`src/xsubj/eeg/preprocess.py`; trial containers and augmentation in `eeg/dataset.py`	cached in `models/xsubj/cache/` on the first run
Video and audio features (4.1)	`src/xsubj/video/clip_features.py` (CLIP), `video/foundation_features.py` (CLAP), `eeg_pipeline/rsa_features.py` (AlexNet, VideoMAE, loudness); loaded by `video/embeddings.py`	`models/xsubj/features/` (versioned; no need to re-extract)
Cross-subject encoders, CLIP vote (4.2)	`src/xsubj/nets/encoders.py`, trained with `src/xsubj/train.py`	`models/xsubj/checkpoints/` `all`, `shallow_s1-s4`, `mlp_s0-s2`, `eegnet_s0`
Ridge votes (4.2)	`src/xsubj/nets/ridge.py`; refitted by `submit.py` on every run (deterministic, seconds)	saved to `checkpoints/ridge`, `ridge_subject` (CLIP) and `ridge_<vote>/` (pooled + one per subject)

Per-subject InfoNCE votes (4.2)	`src/xsubj/infonce_member.py` (training and inference)	`checkpoints/infonce_fc6/`, `infonce_vmae8/`: `sub-XX_s<seed>.pt`, 6 subjects × 3 seeds
Scoring and Sinkhorn (4.3)	`src/xsubj/metrics.py`: `cosine_term()`, `sinkhorn()`	–
Orchestration and output	`src/xsubj/submit.py`; also scores sub-01..04 as a format and accuracy check	`submissions/sub-05_…`, `sub-06_ranked_candidate_list.npy`

10.2 Where each setting lives in `configs/xsubj.yaml`

Key	Meaning (report section)
`preprocess`	40 Hz low-pass, 0–1 s, 100 Hz, mean-response removal, clipping, Euclidean alignment (2)
`model`, `train`	architecture and training of the cross-subject encoders (4.2)
`ridge` (`erp_tmin`, `erp_tmax`, `alpha_scale`)	ridge features 50–600 ms; regularisation × 0.2 (4.2)
`submit.encoders`, `image_weight`	the 9 encoders of the CLIP vote and its weight 0.75 (4.3)
`submit.pooled_weight`	pooled ridge × 0.25 next to the per-subject ridge (4.2)
`submit.extra_votes`	conv3, VideoMAE l8, CLAP and loudness ridge votes with their feature files and weights (4.3)
`submit.infonce_votes`	InfoNCE fc6 and VideoMAE l8 votes, weights and seeds (4.3)
`submit.scoring`	cosine terms and Sinkhorn τ = 0.3 (4.3)

10.3 Not part of the final model

The repository also keeps code from earlier stages of the project; the submission does not use it. In `src/` outside `src/xsubj/`: the organizers' baseline (`baseline_organizer.py`) and first-round experiments (`bench_embeddings.py`, `challenge_data.py`, `eeg_encoder.py`, `eval_matching.py`, `extract_embeddings.py`, `figures.py`, `load_data.py`, `preprocess.py`, `snr_probe.py`, `train_contrastive.py`, `video_embeddings.py`). In `src/xsubj/`: development tools (`eval.py`, `sweep_features.py`, `sweep_preprocess.py`, `predict.py`, `audio_member.py`). Also `configs/default.yaml`, `notebooks/` and the organizers' notebooks in `provided_material/`. The data analysis and RSA of Sections 2–3 are on branch `feature/eeg-preprocessing`.

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